Network Multipliers in Gravity Models

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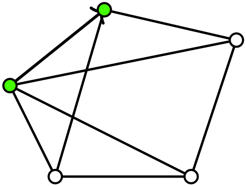
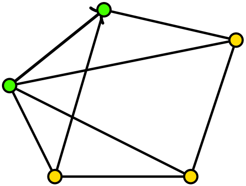
Presentation Slides

- Our paper develops econometric model specificaton and estimation/inference for analyzing forces, signals, and flows within a *system*.
  - "system": subject to connectivity/networks



Without considering the system



With considering the system

- Examples of economic flows: **Trade flows**, migration flows, commuting flows

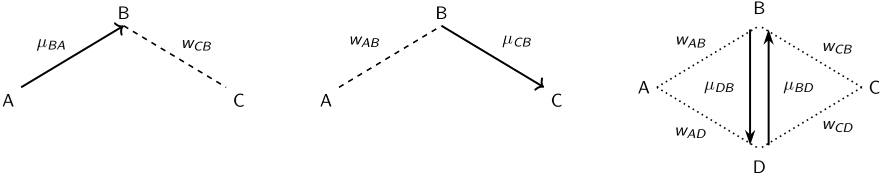
- **Gravity equations** are the workhorse for explaining **trade flows**.
- Specifying **trade costs** is essential.
  - Why? Shape the composition of partners and the quantities exchanged.
  - **Logistics** serve as a fundamental source of economic frictions.
- The *iceberg-cost* specification (Samuelson 1952, 1954)—under which a fraction of the shipped good "melts" in transit—has become the default,
  - enters trade shares multiplicatively (and is therefore tractable).
  - This formulation treats trade costs as largely **exogenous** and unavoidable.

- Question: Do countries' network-leveraging behaviors significantly explain the residuals from iceberg trade-cost models?
  - Yes. Model fit significantly improves: McFadden's R^2 28% (>20%).
- Implication: How do the network spillovers reshape the global trade?
  - How do bilateral trade policy shocks (e.g., tariffs, sanctions, or supply-chain disruptions) propagate through the trade network?
  - How this affects *third-country* trade patterns?
- Some notes
  - Two types of networks
    1. Expected trade flows
    2. Countries' geographic/economic proximities
  - Cross-sectional analysis

1. Introduction

Core Idea

- Case 1 (Cross-destination linkage): Suppose B and C are connected. Suppose A wants to export to C and A knows the expected trade flow from A to B.
- Case 2 (Cross-origin linkage): Suppose A and B are connected. Suppose A wants to export to C and A knows the expected trade flow from B to C.



- Q. How would A account for trade cost exporting to C?
- A1. A would think if it's worth routing through B when exporting to C.
- A2. A would think if it's worth co-shipping with B when exporting to C.
  - The expected trade volume from B to C is important. (If not big, A will less likely leverage B when exporting to C.)
  - If leveraging B, how close with B is also important.

- Countries may leverage their **trade networks** as a *resource* to operate efficient trade costs scheme.
  - (i) expected trade flows and (ii) the geographic/economic ties

$\Rightarrow$ *Interdependent trade flows* via specifying a cost function.

  - Trade costs emerge *endogenously* from trade networks, rather than remaining exogenous.
    - ★ i.e., a country's trade costs *depend* on the expected trade flows of others with countries' connectivities.
  - This feature is captured by the **spatial autoregressive (SAR)** framework.

- <u>Note.</u> The iceberg costs are a special case of our framework.

Role of **Spatial Autoregressive (SAR)** Model

- SAR represents interdependence: $y_i = w_{i1}y_1 + w_{i2}y_2 + \cdots + w_{in}y_n + \ldots$
  - y_i: i's outcome, $w_{i1}y_1 + w_{i2}y_2 + \cdots + w_{in}y_n$: neighbors' outcome
  - Can accommodate endogenous trade costs scheme
- SAR represents *network-induced* pair-specific heterogeneity
  - Multilateral resistance traditionally reflects average trade barriers across *all* partners, modeled by individual fixed effects.
  - Our framework *extends* this concept by allowing for interdependence within these terms.
  - The resulting *pair-specific heterogeneity* is thus inherently relational, manifesting at the *pair* level.

Our paper is about **Econometric model specification.**

1. [Why] We **endogenize trade costs** through the lens of trade networks as a resource in operating logistics-related trade costs.
    - Improve model fit by reducing residuals through econometric model specification.
    - Structural gravity equation for trade flows/iceberg trade cost specification
        - ★ Krugman (1995), Eaton and Kortum (2002 *Ecta*), Anderson and van Wincoop (2003 *AER*), Arkolakis et al. (2012 *AER*), Tyazhelnikov and Romalis (2024 *JIE*)

2-1. [How] From this motivation, we characterize **interdependent trade flows**.
- Interdependence from different motivations
    - ★ Allen et al. (2020 *JPE*), Lind and Ramondo (2023 *AER*)
- Spatial autoregressive framework
    - ★ Cliff and Ord (1995), Ord (1975 *JASA*), Lee (2004 *Ecta*, 2007 *JoE*)
    - ★ **LeSage and Pace (2008 *JRS*)**
    - ★ Behrens et al. (2012 *JAE*), Pesaran and Yang (2021 *JoE*), Jin et al. (2023 *ER*), Jeong et al. (2023, *EL*), Jeong and Lee (2024 *JoE*)
- For estimation, we developed the Poisson pseudo-maximum likelihood estimation with network effects.
- For inference, spatial HAC estimator.

2-2. [Innovation] Efficient computation method

- Network multiplier matrix: need to calculate an *inverse* of a huge matrix
  - ★ Makes computational cost heavy
- Based on the spectral decomposition of a network matrix, our algorithm
  - ★ yields a linear equation of the network multiplied quantities;
  - ★ and eventually leads to a closed form solution.
  - ★ i.e., We don't have to calculate an inverse of the matrix at all!
- Computation time comparison

n	$N = n^2$	Default (A)	Ours (B)	A/B
9	81	0.5668	0.3518	1.6112
25	625	12.3862	0.3967	31.2265
49	2401	1252.0077	3.9725	315.1687
64	4096	9465.7145	9.2816	1019.8380

3. Empirical analysis
   - Two phases of global trade
     1. Phase 1 (1997, active NAFTA implementation)
     2. Phase 2 (2016, expansion of global value chains)
   - Network analysis
     - ★ Trade networks became more interconnected and denser, deviating from a bipartite structure
   - Estimation results: Complimentarity under NAFTA and mature global value chains, respectively.
   - Counterfactual experiments and their policy implications
     - ★ How conflicts between major economies (e.g., US–China trade war) spill over to third countries and reshape global trade structures

1. Introduction

Outline

1. Specification
    - Microfoundation
    - Econometric Model specification
2. Estimation/Inference
    - Poisson Pseudo Maximum Likelihood Estimation Under Network Influence
    - Asymptotic Distribution - Main and fixed-effect parameters
3. Monte Carlo Simulations
4. Empirical Application
    - Basic setup/Network statistics
    - Estimation results
    - Counterfactual experiments

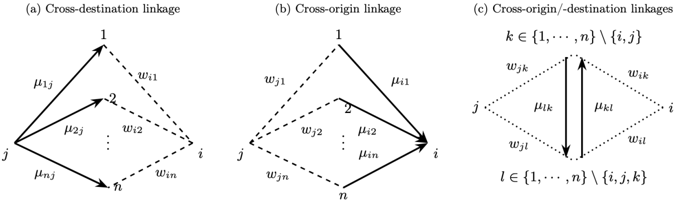
Assumption (Spatial Setting)

Each $i \in \{1, \cdots, n\}$ is in a d-dimensional space $\mathcal{D}_n \subset \mathcal{D}$, where $\mathcal{D}$ denotes a set of all potential locations in $\mathbb{R}^d$.
We assume $\lim_{n\to\infty} \#(\mathcal{D}_n) = \infty$ and $\min_{i \neq j} d(l(i), l(j)) \geq 1$, where $\#(\mathcal{D}_n)$ is the cardinality of $\mathcal{D}_n$, $l : i \mapsto l(i) \in \mathcal{D}$ stands for an injective location function, and $d(l(i), l(j))$ is a distance between i and j.

- Trade flow from j to i, y_{ij}, is generated by two locations, origin j and destination i.

$\Rightarrow$ There are n locations in a sample, so $N = n^2$ flow outcomes are observed.

- Each w_{ij} denotes proximity between i and j.
  - Note. In practice, W can be constructed using historical trade flows.
    - ★ w_{ij} = long-run relationship between i and j
  - Free from the construction issue of W, thanks to the nature of network data.



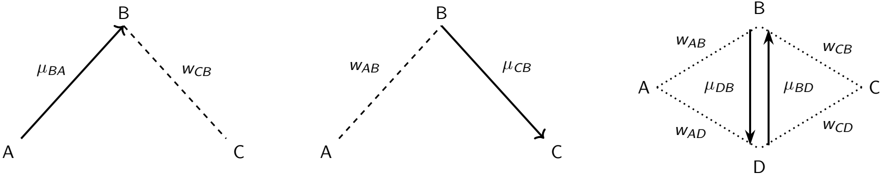
- **Stage 1 (partner (hub) selection):** For each ν, i forms a probability distribution over potential partners $k \in \{1, \cdots, n\} \setminus \{i\}$, and j forms a probability distribution over $l \in \{1, \cdots, n\} \setminus \{j\}$:

$$\Pr(k \text{ is } i\text{'s partner at } \nu) = w_{ik}^{d} \text{ and } \Pr(l \text{ is } j\text{'s partner at } \nu) = w_{jl}^{o}.$$

  - For empirical application, we adopt a single set of proximity weights and impose $W^d = W^o = W$.

2. Specification

Microfoundation



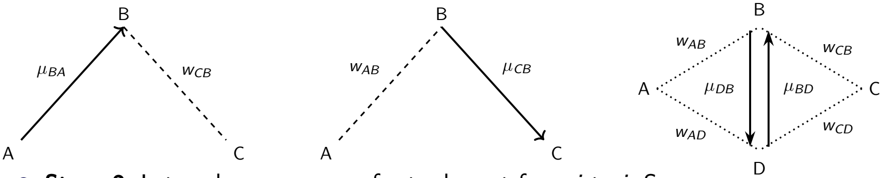
- **Stage 2:** Let π_{CA} be a measure of a trade cost from A to C. Suppose

$$\pi_{CA}(\boldsymbol{\mu}) = \Bigg(\underbrace{\left(\mu_{BA}^{w_{CB}}\right)^{\tilde{\lambda}_d}}_{\substack{\text{outflows from } A \\ \text{cross-destination linkage}}} \times \underbrace{\left(\mu_{CB}^{w_{AB}}\right)^{\tilde{\lambda}_o}}_{\substack{\text{inflows to } C \\ \text{cross-origin linkage}}} \times \underbrace{\left(\cdots\left(\mu_{DB}^{w_{AB}w_{CD}}\right)\cdot\left(\mu_{BD}^{w_{AD}w_{CB}}\right)\cdots\right)^{\tilde{\lambda}_w}}_{\substack{\text{flows among third-party units} \\ \text{cross-origin and cross-destination linkage}}} \Bigg)^{-1} \times \underbrace{D_{CA}^{\tilde{\beta}}}_{\substack{\text{bilateral} \\ \text{characteristics}}}$$

  - μ_{ij}: expected trade flow y_{ij}
  - $\tilde{\lambda}_d$, $\tilde{\lambda}_o$, $\tilde{\lambda}_w$, $\tilde{\beta}$: structural parameters

2. Specification

Microfoundation



- **Stage 2:** Let π_{ij} be a measure of a trade cost from j to i. Suppose

$$\pi_{ij}(\boldsymbol{\mu}) = \left(\underbrace{\left(\prod_{k=1}^{n} \mu_{kj}^{w_{ik}}\right)^{\tilde{\lambda}_d}}_{\substack{\text{outflows from } j \\ \text{cross-destination linkage}}} \times \underbrace{\left(\prod_{l=1}^{n} \mu_{il}^{w_{jl}}\right)^{\tilde{\lambda}_o}}_{\substack{\text{inflows to } i \\ \text{cross-origin linkage}}} \times \underbrace{\left(\prod_{k,l=1}^{n} \mu_{kl}^{w_{ik}w_{jl}}\right)^{\tilde{\lambda}_w}}_{\substack{\text{flows among third-party units} \\ \text{cross-origin and cross-destination linkage}}} \times \underbrace{D_{ij}^{\tilde{\beta}}}_{\substack{\text{bilateral} \\ \text{characteristics}}} \right)^{-1}$$

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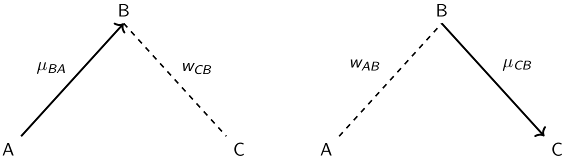

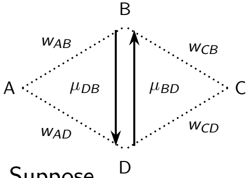

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$$\times \underbrace{\underbrace{D_{ij}^{\tilde{\beta}}}_{\substack{\text{bilateral} \\ \text{characteristics}}}}_{=:\pi^{+}_{ij} \text{ (exogenous part)}}$$

Full derivation

Stage 3: Given $\pi_{ij}(\boldsymbol{\mu})$ from **Stage 2**, the optimal trade flows are determined as in Anderson and van Wincoop (2003).

- At equilibrium, the total trade flow from exporter j to importer i is

$$\mu_{ij}^* = \frac{G_i G_j}{G^W} \left(\frac{\pi_{ij}(\boldsymbol{\mu}^*)}{\Pi_j(\boldsymbol{\mu}^*) P_i(\boldsymbol{\mu}^*)} \right)^{1-\varrho},$$

  - G_i: i's budget (exogenously given)
  - G^W: World budget
  - $\Pi_j(\boldsymbol{\mu})$ and $P_i(\boldsymbol{\mu})$: multilateral resistance terms at $\boldsymbol{\mu}$.
  - ϱ: CES elasticity

- Plugging $\pi_{ij}(\boldsymbol{\mu})$ from **Stage 2**,

$$\mu_{ij} = \underbrace{(\pi_{ij}^e(\boldsymbol{\mu}))^{1-\varrho}}_{\text{explicitly endogenous term}} \cdot \underbrace{P_i^{\varrho-1}(\boldsymbol{\mu})\Pi_j^{\varrho-1}(\boldsymbol{\mu})}_{\text{implicitly endogenous term}} \cdot \underbrace{G_i G_j (G^W)^{-1} (\pi_{ij}^{+})^{1-\varrho}}_{\text{purely exogenous term}},$$

  - $(\pi_{ij}^e(\boldsymbol{\mu}))^{1-\varrho} = \left(\prod_{k=1}^n \mu_{kj}^{w_{ik}}\right)^{\lambda_d} \left(\prod_{l=1}^n \mu_{il}^{w_{jl}}\right)^{\lambda_o} \left(\prod_{k,l=1}^n \mu_{kl}^{w_{ik} w_{jl}}\right)^{\lambda_w}$,
  where $\lambda_d = (\varrho - 1)\tilde{\lambda}_d$, $\lambda_o = (\varrho - 1)\tilde{\lambda}_o$, $\lambda_w = (\varrho - 1)\tilde{\lambda}_w$.

Assumption (Equilibrium uniqueness (i))

$\rho_{\text{spec}}(\mathbf{A}) < 1$, where $\mathbf{A}$ is an aggregate network matrix defined by

$$\mathbf{A} := \underbrace{\lambda_d(I_n \otimes W)}_{\substack{\text{cross-destination}\\ \text{linkage}}} + \underbrace{\lambda_o(W \otimes I_n)}_{\substack{\text{cross-origin}\\ \text{linkage}}} + \underbrace{\lambda_w(W \otimes W)}_{\substack{\text{cross-origin}\\ \text{and -destination linkages}}},$$

with $\lambda_d = (\varrho - 1)\tilde{\lambda}_d$, $\lambda_o = (\varrho - 1)\tilde{\lambda}_o$, $\lambda_w = (\varrho - 1)\tilde{\lambda}_w$.

- Semi-reduced form: Under (i) and (ii), there is a unique $\boldsymbol{\mu}^*$ satisfying

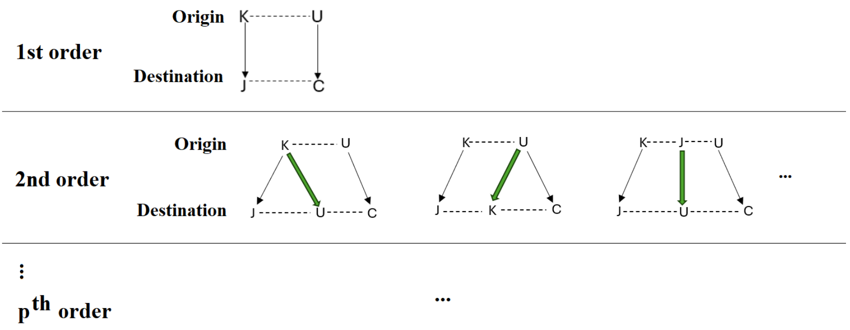
$$\mu_{ij}^* = \exp\left(\sum_{k,l=1}^{n} s_{ij,kl}\left(x_{kl}'\beta + \alpha_l(\boldsymbol{\mu}^*) + \eta_k(\boldsymbol{\mu}^*)\right)\right), \text{ for } i,j = 1,\cdots,n,$$

where $x_{kl} = (\ln(D_{kl,1}),\cdots,\ln(D_{kl,K}))'$ and $\beta = (\beta_1,\cdots,\beta_K)'$ with $\beta_k = (1-\varrho)\tilde{\beta}_k$.

  - $\mathbf{S}^{-1}$: Network multiplier matrix
    - $s_{ij,kl} \equiv ((j-1)n + i, (l-1)n + k)$-element of $\mathbf{S}^{-1}$.
    - $\mathbf{S} = I_N - \mathbf{A}$: Network SAR operator (LeSage and Pace (2008) *JRS*)
  - *Network Multilateral Resistance* fixed effects: $\alpha_l(\boldsymbol{\mu}^*)$ and $\eta_k(\boldsymbol{\mu}^*)$

<u>Remark.</u> This specification encompasses all intermediate flows that carry the signal $s_{ij,kl}$ between the flows (k, l) and (i, j) by p-th order, $p \in \mathbb{N}$.

Eg. The signal $s_{KJ,UC}$ between $(Korea, Japan)$ and $(US, China)$ is carried by inifinitely many intermediate flows as....



Assumption (Equilibrium uniqueness (ii))

$\boldsymbol{\mu}^*$ satisfies the following condition:

$$\sup_{i,j} \sum_{k,l=1}^{n} \left| \sum_{p,q=1}^{n} s_{ij,pq} \left(\frac{\partial \left(\alpha_q(\boldsymbol{\mu}) + \eta_p(\boldsymbol{\mu}) \right)}{\partial \ln(\mu_{kl})} \right) \right| < 1.$$

In words, a small change of μ_{kl} does *not* yield dramatic changes in the network fixed effects, which include the multilateral resistance terms.

- $\alpha_q(\boldsymbol{\mu}) = \text{const.} + \ln(G_q) + \ln\left(\Pi_q^{\varrho-1}(\boldsymbol{\mu})\right)$ for $q = 1, \cdots, n$,
- $\eta_p(\boldsymbol{\mu}) = \text{const.} + \ln(G_p) + \ln\left(P_p^{\varrho-1}(\boldsymbol{\mu})\right)$ for $p = 1, \cdots, n$,
- $\Pi_j(\boldsymbol{\mu}) = \left(\sum_{i=1}^{n} \frac{G_i}{G^W} \left(\frac{\pi_{ij}(\boldsymbol{\mu})}{P_i(\boldsymbol{\mu})} \right)^{1-\varrho} \right)^{\frac{1}{1-\varrho}}$: Price index for origin j
- $P_i(\boldsymbol{\mu}) = \left(\sum_{j=1}^{n} \frac{G_j}{G^W} \left(\frac{\pi_{ij}(\boldsymbol{\mu})}{\Pi_j(\boldsymbol{\mu})} \right)^{1-\varrho} \right)^{\frac{1}{1-\varrho}}$: Price index for destination i

- From the semi-reduced form, the true data-generating process can be specified by

$$y_{ij} = \mu_{ij}^0 \times \xi_{ij}, \text{ where } \mu_{ij}^0 = \exp\left(\sum_{k,l=1}^{n} s_{ij,kl}\left(x_{kl}'\beta^0 + \alpha_l^0 + \eta_k^0\right)\right),$$

  - $\mu_{ij}^0 = \mathbb{E}(y_{ij}|\mathbf{z})$: Economic Model; ξ_{ij}: Error satisfying $\mathbb{E}(\xi_{ij}|\mathbf{z}) = 1$
  - $\mathbf{z}$ stands for a vector of exogenous characteristics
  - $u_{ij} = \mu_{ij}^0(\xi_{ij} - 1)$: additive error, $\mathbb{E}(u_{ij}|\mathbf{z}) = 0$
  - $\lambda^0 = \left(\lambda_d^0, \lambda_o^0, \lambda_w^0\right)'$ denotes a vector of the true network influence parameters
  - $\beta^0 = \left(\beta_1^0, \cdots, \beta_K^0\right)'$ is the true parameter for x_{kl}
  - α_j^0 and η_i^0 are the true origin- and destination- fixed effects, respectively.

- Remarks
  - The purpose of this model is to identify λ^0, β^0, $\alpha_1^0, \cdots, \alpha_n^0$, $\eta_1^0, \cdots,$ and η_n^0.
  - μ_{ij}^0 $(i,j = 1, \cdots, n)$ can be identified by the semi-reduced form.

Estimation

3. Estimation

Poisson Pseudo Maximum Likelihood Estimation

- **Poisson pseudo maximum likelihood (PPML) estimator** (Gourieroux et al. 1984 *Ecta*) has become the *standard* since Santos Silva and Tenreyro (2006 *REStat*).
- Pseudo log-likelihood

$$\ell_N(\theta, \phi) = \sum_{i,j=1}^{n} \Big(-\mu_{ij}(\theta, \phi) + y_{ij} \ln \mu_{ij}(\theta, \phi) \Big) - \ln y_{ij}! - \frac{1}{2} \left(\sum_{j=1}^{n} \alpha_j - \sum_{i=1}^{n} \eta_i \right)^2,$$

where $\phi := (\boldsymbol{\alpha}', \boldsymbol{\eta}')'$.

  - PPMLE: $\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \ell_N(\boldsymbol{\theta})$, where $\boldsymbol{\theta} = (\theta', \phi')'$
- First-order conditions: $\partial_\theta \ell_N(\boldsymbol{\theta}) = \sum_{i,j=1}^{n} \partial_\theta \tilde{\mu}_{ij}(\boldsymbol{\theta})(y_{ij} - \exp(\cdot))$
  - Recall $u_{ij} = y_{ij} - \mu_{ij}^0$.
  - Since $\mathbb{E} u_{ij} | \mathbf{z} = 0$, $\mathbb{E}(h(\mathbf{z}; \boldsymbol{\theta}) u_{ij}) = 0$ for any measurable functions h.
  - This implies as long as $\mu_{ij}^0 = \mathbb{E}(y_{ij} | \mathbf{z}) = \exp(\cdot)$, the FOCs correspond to moment conditions.
- Only the first conditional moment restriction is required.
  - The Poisson distribution need not hold. (Poisson *pseudo* MLE)
  - Can accommodate many zero ys $\Rightarrow$ No need to have $\ln(y + 1)$

3. Estimation

PPMLE - Efficient computation algorithm for $\mathbf{S}^{-1}(\lambda)\mathbf{Z}(\boldsymbol{\theta})$

- Main issue here
  - Need to compute $\mathbf{S}^{-1}(\lambda)\mathbf{Z}(\boldsymbol{\theta})$ for each $\boldsymbol{\theta}$.
  - Do we need to invert $\mathbf{S}(\lambda)$?
  - When there are $n = 150$ countries, $\mathbf{S}(\lambda)$ is an $150^2 \times 150^2$ matrix ($506, 250, 000$ elements!)
- Core idea
  - Utilize the structure of $\mathbf{A} = \lambda_d(I_n \otimes W) + \lambda_o(W \otimes I_n) + \lambda_w(W \otimes W)$ ($\mathbf{S}(\lambda) = I_N - \mathbf{A}(\lambda)$)
  - Utilize $W = QDQ^{-1}$ (obtain this decomposition before estimation).
    - ★ Q is the eigenvector basis of W
    - ★ D is the diagonal matrix consisting of eigenvalues of W

Recall $\mathbf{S}^{-1} = (I_N - \mathbf{A})^{-1}$, where $\mathbf{A} = \lambda_d(I_n \otimes W) + \lambda_o(W \otimes I_n) + \lambda_w(W \otimes W)$.

- For the invertibility of **S**, we assume $\rho_{\mathbf{spec}}(\mathbf{A}) < 1$.
- We have $\rho_{\mathbf{spec}}(\mathbf{A}) = \max\{b(1,1), b(1,\varphi_{\mathbf{min}}), b(\varphi_{\mathbf{min}},1), b(\varphi_{\mathbf{min}},\varphi_{\mathbf{min}})\}$, where
  - $b(1,1) = \lambda_d + \lambda_o + \lambda_w,\ b(1,\varphi_{\mathbf{min}}) = \lambda_d\varphi_{\mathbf{min}} + \lambda_o + \lambda_w\varphi_{\mathbf{min}},$
    $b(\varphi_{\mathbf{min}},1) = \lambda_d + \lambda_o\varphi_{\mathbf{min}} + \lambda_w\varphi_{\mathbf{min}},$
    $b(\varphi_{\mathbf{min}},\varphi_{\mathbf{min}}) = \lambda_d\varphi_{\mathbf{min}} + \lambda_o\varphi_{\mathbf{min}} + \lambda_w\varphi^2_{\mathbf{min}}.$
- Why? $I_n \otimes W,\ W \otimes I_n,$ and $W \otimes W$ share the same eigenvector basis:

$$
\begin{aligned}
(I_n \otimes W)(q_i \otimes q_j) &= q_i \otimes Wq_j = q_i \otimes \varphi_j q_j = \varphi_j(q_i \otimes q_j)\\
(W \otimes I_n)(q_i \otimes q_j) &= Wq_i \otimes q_j = \varphi_i q_i \otimes q_j = \varphi_i(q_i \otimes q_j)\\
(W \otimes W)(q_i \otimes q_j) &= Wq_i \otimes Wq_j = (\varphi_i q_i) \otimes (\varphi_j q_j) = (\varphi_j\varphi_j)(q_i \otimes q_j),
\end{aligned}
$$

  where Q is some eigenvector basis of W and q_i is the i$^{\text{th}}$ column vector of Q.

  - Thus,

$$
\begin{aligned}
\mathbf{A}(q_i \otimes q_j) &= (\lambda_d(I_n \otimes W) + \lambda_o(W \otimes I_n) + \lambda_w(W \otimes W))(q_i \otimes q_j)\\
&= (\lambda_d\varphi_j + \lambda_o\varphi_i + \lambda_w\varphi_i\varphi_j)(q_i \otimes q_j), \quad i,j = 1,\ldots,n.
\end{aligned}
$$

  - Note that the eigenvalue of **A** is a bilinear map:
    $b(\varphi_i,\varphi_j) = \lambda_d\varphi_j + \lambda_o\varphi_i + \lambda_w\varphi_i\varphi_j$ for $(\varphi_i,\varphi_j) \in [\varphi_{\mathbf{min}},1]^2$.

- Recall $\mathrm{A}(\lambda)(q_i \otimes q_j) = (\lambda_d\varphi_j + \lambda_o\varphi_i + \lambda_w\varphi_i\varphi_j)(q_i \otimes q_j)$ for $i, j = 1, \cdots, n$.
- We want to obtain the matrix fixed-point $T(\boldsymbol{\theta})$ satisfying $\mathrm{vec}\,(T(\boldsymbol{\theta})) = \mathbf{S}^{-1}(\lambda)\mathbf{Z}(\boldsymbol{\theta})$ for each $\boldsymbol{\theta}$.
  - $Z^{\mathrm{mat}}(\boldsymbol{\theta})$ is such that $\mathbf{Z}(\boldsymbol{\theta}) = \mathrm{vec}\,(Z^{\mathrm{mat}}(\boldsymbol{\theta}))$.
- Note that Q and D are invariant in the estimation procedure. Hence, we have

$$\widetilde{Z}^{\mathrm{mat}}(\boldsymbol{\theta}) = \widetilde{T}(\boldsymbol{\theta}) - \lambda_d D\widetilde{T}(\boldsymbol{\theta}) - \lambda_o\widetilde{T}(\boldsymbol{\theta})D - \lambda_w D\widetilde{T}(\boldsymbol{\theta})D,$$

  - $\widetilde{Z}^{\mathrm{mat}}(\boldsymbol{\theta}) = Q^{-1}Z^{\mathrm{mat}}(\boldsymbol{\theta})Q^{-1\prime}$
  - $\widetilde{T}(\boldsymbol{\theta}) = Q^{-1}T(\boldsymbol{\theta})Q^{-1\prime}$
- This implies

$$(\widetilde{T}(\boldsymbol{\theta}))_{ij} = \frac{(\widetilde{Z}^{\mathrm{mat}}(\boldsymbol{\theta}))_{ij}}{1 - \lambda_d\varphi_i - \lambda_o\varphi_j - \lambda_w\varphi_i\varphi_j}$$

  by $\mathrm{A}(\lambda)(q_i \otimes q_j) = (\lambda_d\varphi_j + \lambda_o\varphi_i + \lambda_w\varphi_i\varphi_j)(q_i \otimes q_j)$ for $i, j = 1, \cdots, n$.
  - Then, we can easily recover $T(\boldsymbol{\theta}) = Q\widetilde{T}(\boldsymbol{\theta})Q'$.

3. Estimation

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49	2401	1252.0077	3.9725	315.1687
64	4096	9465.7145	9.2816	1019.8380

- Default (A) maximizes the log-likelihood based on the direct computation of $\mathbf{S}^{-1}(\lambda)$.
- Ours (B) maximizes the log-likelihood based on our suggested method.
- When $n \simeq 150$, our proposed method is **more than four orders of magnitude faster**, rendering maximum likelihood estimation computationally feasible even for moderately large n.

Assumption (Identification)

Let Θ_θ denote a compact parameter space of θ and $\boldsymbol{\Phi}$ represent a parameter space of ϕ.
(i) Assume $\liminf_{n\to\infty} \inf_{\phi\in\boldsymbol{\Phi}} \varphi_{\min}\left(\frac{1}{n}\mathbf{D}'\mathbf{S}^{-1\prime}(\lambda)\mathrm{Diag}\left(\boldsymbol{\mu}(\boldsymbol{\theta})\right)\mathbf{S}^{-1}(\lambda)\mathbf{D} + \text{other terms}\right) > 0$ for each $\theta \in \Theta_\theta$. Then, $\hat{\phi}(\theta) = \mathrm{argmax}_{\phi\in\boldsymbol{\Phi}} \ell_N(\theta, \phi)$ is unique for each $\theta \in \Theta_\theta$ and for a large n.
(ii) For each $\theta \in \Theta_\theta$, let

$$\widehat{\mathbf{H}}(\theta) = \frac{1}{N}\widehat{\mathbf{G}}'(\theta)\mathbf{S}^{-1\prime}(\lambda)\mathrm{Diag}\left(\widehat{\boldsymbol{\mu}}(\theta)\right)\mathbf{S}^{-1}(\lambda)\widehat{\mathbf{G}}(\theta) + \text{other terms},$$

where $\widehat{\mathbf{G}}(\theta) = \mathbf{G}\left(\theta, \hat{\phi}(\theta)\right)$ and $\widehat{\boldsymbol{\mu}}(\theta) = \boldsymbol{\mu}(\theta, \hat{\phi}(\theta))$ for each $\theta \in \Theta_\theta$.
Assume $\liminf_{n\to\infty} \inf_{\theta\in\Theta_\theta} \varphi_{\min}\left(\widehat{\mathbf{H}}(\theta)\right) > 0$.
$\Rightarrow$ Uniqueness of $\theta^0 = \mathrm{argmax}_{\theta\in\Theta_\theta} \ell_\infty(\theta, \phi(\theta))$, and consequently, $\phi^0 = \phi(\theta^0)$.

- Notations
  - $\mathbf{D} = [\mathrm{I}_n \otimes l_n, l_n \otimes \mathrm{I}_n]$ is an $N \times 2n$ matrix for dummy variables
  - $\mathbf{G}(\boldsymbol{\theta}) = \left[(I_n \otimes W)\mathbf{S}^{-1}(\lambda)\mathbf{Z}(\boldsymbol{\theta}), (W \otimes I_n)\mathbf{S}^{-1}(\lambda)\mathbf{Z}(\boldsymbol{\theta}), (W \otimes W)\mathbf{S}^{-1}(\lambda)\mathbf{Z}(\boldsymbol{\theta}), \mathbf{X}\right]$ and $\mathbf{Z}(\boldsymbol{\theta}) = \mathbf{X}\beta + \boldsymbol{\alpha} \otimes l_n + l_n \otimes \boldsymbol{\eta}$
  - $\boldsymbol{\mu}(\boldsymbol{\theta}) = \left(\exp\left(\tilde{\mu}_{11}(\boldsymbol{\theta})\right), \cdots, \exp\left(\tilde{\mu}_{n1}(\boldsymbol{\theta})\right), \cdots, \exp\left(\tilde{\mu}_{1n}(\boldsymbol{\theta})\right), \cdots, \exp\left(\tilde{\mu}_{nn}(\boldsymbol{\theta})\right)\right)$ with $\tilde{\mu}_{ij}(\boldsymbol{\theta}) = \sum_{k,l=1}^{n} s_{ij,kl}(\lambda)\left(x_{kl}'\beta + \alpha_l + \eta_k\right)$

Theorem

Under some regularity conditions,

$$\sqrt{N}\left(\hat{\theta}-\theta^0\right) \xrightarrow{d} N\left(0, \mathbf{\Sigma}_\theta^{-1}\mathbf{\Omega}_\theta\mathbf{\Sigma}_\theta^{-1}\right) \text{ as } n \to \infty,$$

where $\mathbf{\Sigma}_\theta = \operatorname{plim}_{n\to\infty} \mathbf{\Sigma}_{\theta,N}$, $\mathbf{\Omega}_\theta = \operatorname{plim}_{n\to\infty} \mathbf{\Omega}_{\theta,N}$.

- Here,

$$\mathbf{\Sigma}_{\theta,N} = \frac{1}{N}\mathbf{G}'\mathbf{S}^{-1\prime}\mathrm{Diag}(\boldsymbol{\mu})\mathbf{M}_{\mathbf{D}}\mathbf{S}^{-1}\mathbf{G},$$

and

$$\mathbf{\Omega}_{\theta,N} = \frac{1}{N}\mathbf{G}'\mathbf{S}^{-1\prime}\mathbf{M}_{\mathbf{D}}'\mathbb{E}\left(\mathbf{u}\mathbf{u}'\right)\mathbf{M}_{\mathbf{D}}\mathbf{S}^{-1}\mathbf{G}.$$

  - $\mathbf{M}_{\mathbf{D}} = I_N - \mathbf{P}_{\mathbf{D}}\mathrm{Diag}(\boldsymbol{\mu})$ and $\mathbf{P}_{\mathbf{D}} = \mathbf{S}^{-1}\mathbf{D}(\widetilde{\mathbf{D}'\mathbf{D}})^{-1}\mathbf{D}'\mathbf{S}^{-1\prime}$ with $\widetilde{\mathbf{D}'\mathbf{D}} = \mathbf{D}'\mathbf{S}^{-1\prime}\mathrm{Diag}(\boldsymbol{\mu})\mathbf{S}^{-1}\mathbf{D} - \mathbf{H}^{\phi\phi}$

- No asymptotic bias

Theorem (Fixed-effect estimators)

Under some regularity conditions,

$$\sqrt{n}\left(\widehat{\alpha}_j - \alpha_j^0\right) \xrightarrow{d} N(0, \lim_{n\to\infty} e_{2n,j}' \boldsymbol{V}_{\phi,N} e_{2n,j}), \text{ and}$$

$$\sqrt{n}\left(\widehat{\eta}_i - \eta_i^0\right) \xrightarrow{d} N(0, \lim_{n\to\infty} e_{2n,n+i}' \boldsymbol{V}_{\phi,N} e_{2n,n+i})$$

as $n \to \infty$.

- Using $\{\widehat{\alpha}_j\}$ and $\{\widehat{\eta}_i\}$, we can identify/estimate γ_o^0 and γ_d^0.
- Note that

$$\mathbf{V}_{\phi,N} = n\left(\widetilde{\mathbf{D}'\mathbf{D}}\right)^{-1}\mathbf{D}'\mathbf{S}^{-1\prime}\mathbf{M}_\phi'\mathbb{E}(\mathbf{u}\mathbf{u}')\mathbf{M}_\phi\mathbf{S}^{-1}\mathbf{D}\left(\widetilde{\mathbf{D}'\mathbf{D}}\right)^{-1},$$

  where

$$\mathbf{M}_\phi = I_N - \mathbf{M}_\mathbf{D}\mathbf{S}^{-1}\mathbf{G}\left(\mathbf{G}'\mathbf{S}^{-1\prime}\mathrm{Diag}(\boldsymbol{\mu})\mathbf{M}_\mathbf{D}\mathbf{S}^{-1}\mathbf{G}\right)^{-1}\mathbf{G}'\mathbf{S}^{-1\prime}\mathrm{Diag}(\boldsymbol{\mu}).$$

- $e_{2n,j}$ denotes the $2n$-dimensional unit vector with its j-th element equal to 1 and all other elements equal to 0.

3. Estimation

PPMLE - Asymptotic Distribution

- $\mathbf{\Omega}_\theta$ contains $\mathbb{E}(\mathbf{uu}')$, where $\mathbf{u} = (u_{11}, \cdots, u_{n1}, \cdots, u_{1n}, \cdots, u_{nn})'$.

Theorem (Variance estimators)

Let d_N denote the pre-specified threshold distance and $d^*_{ij,kl}$ the distance between ij and kl. Let the spatial HAC estimator and the infeasible spatial HAC estiamator, respectively,

$$\widehat{\mathbf{\Omega}}_{\theta,N} = \frac{1}{N} \sum_{i,j,k,l=1}^{n} b_{ij} c'_{kl} \hat{u}_{ij} \hat{u}_{kl} \cdot \mathrm{K}\left(\frac{d^*_{ij,kl}}{d_N}\right) \text{ for some } b_{ij} \text{ and } c_{kl},$$

$$\widetilde{\mathbf{\Omega}}_{\theta,N} = \frac{1}{N} \sum_{i,j,k,l=1}^{n} b_{ij} c'_{kl} u_{ij} u_{kl} \cdot \mathrm{K}\left(\frac{d^*_{ij,kl}}{d_N}\right) \text{ for some } b_{ij} \text{ and } c_{kl}.$$

Under some regularity conditions,

(i) (Variance) $\lim_{n\to\infty} \frac{N}{\mathbb{E}(deg^*)} \mathrm{Var}\left(vec\left(\widetilde{\mathbf{\Omega}}_{\theta,N}\right)\right) = \overline{\mathrm{K}}(1+C)(\mathbf{\Omega}_\theta \otimes \mathbf{\Omega}_\theta)$;

(ii) (Bias) $\lim_{n\to\infty} d_N^q \left(\mathbb{E}\left(\widetilde{\mathbf{\Omega}}_{\theta,N}\right) - \mathbf{\Omega}_{\theta,N}\right) = -K_q \mathbf{\Omega}_\theta^{(q)}$;

(iii) If $0 < \lim_{n\to\infty} \frac{d_N^{2q}\mathbb{E}(deg^*)}{N} < \infty$, $\sqrt{\frac{N}{\mathbb{E}(deg^*)}} \left(\widehat{\mathbf{\Omega}}_{\theta,N} - \mathbf{\Omega}_{\theta,N}\right) = O_p(1)$,

$\sqrt{\frac{N}{\mathbb{E}(deg^*)}} \left(\widehat{\mathbf{\Omega}}_{\theta,N} - \widetilde{\mathbf{\Omega}}_{\theta,N}\right) = O_p(1)$.

Monte Carlo Simulations

4. Monte Carlo Simulations

- We generate the data based on the structural gravity model.
  - Basic setting
  - $n = 49$, $\lambda_d = 0.2$, $\lambda_o = 0.2$, $\lambda_w = 0.1$, $\beta_1 = 0.6$, and $\beta_2 = 0.2$
  - W is row-normalized, constructed by supposing two influential units.

- Results

	λ_d	λ_o	λ_w	β_1	β_2	α_{49}	η_{49}
Empirical bias	0.0123	-0.0004	-0.0163	-0.0011	-0.0001	0.0131	0.0044
Empirical STD	0.0266	0.0260	0.0355	0.0137	0.0130	0.0845	0.0850
s.e.	0.0229	0.0232	0.0331	0.0120	0.0118	0.0297	0.0228
CP	0.9330	0.9340	0.9280	0.8970	0.9180	0.8630	0.8910

  - Among the four possible kernel choices (Bartlett, Parzen, Tukey–Hanning, QS), Parzen performs best.
  - Among the three distance measures, $d^{*,2}_{ij,kl}$ performs best.
    - ★ $d^{*,1}_{ij,kl} = D_{ik} + D_{jl}$
    - ★ $d^{*,2}_{ij,kl} = \left(D^2_{ik} + D^2_{jl}\right)^{\frac{1}{2}}$
    - ★ $d^{*,\infty}_{ij,kl} = \max\left(D_{ik}, D_{jl}\right)$

Empirical Application

- We suspect different network structures across the following phases:
  - Phase 1: 1997, active NAFTA implementation
  - Phase 2: 2016, expansion of global supply chains.
- Data
  - World trade flows: the Center for International Data at UC Davis
  - Distance, border, legal, language, colony, currency, islands, landlock from Helpman et al. (2008), Chen et al. (2021)
  - FTA from WTO data
  - Due to the data availability, the 142 and 147 countries by phase, respectively, are included.
- Countries' connectivity matrix: Average total trade flows over the recent past years
  - $w_{ij}^{\text{Phase}} = \frac{\widetilde{w}_{ij}^{\textbf{Phase}}}{\sum_{k=\mathbf{1}}^{n} \widetilde{w}_{ik}^{\textbf{Phase}}}$, where $\widetilde{w}_{ij}^{\text{Phase}} = \frac{\mathbf{1}}{\#(\mathcal{T}^{\textbf{Phase}})} \sum_{t \in \mathcal{T}^{\textbf{Phase}}} \left(y_{ij,t} + y_{ji,t}\right)$
  - e.g., $\mathcal{T}^{\text{Phase}=\mathbf{1}} = \{1987, 1988, \cdots, 1996\}$
  - W is undirected and row-normalized, with its diagonal elements being zero.

5. Empirical Application

Network Statistics

	linear-in-means	Bipartite	Phase 1	Phase 2
degree (deg)	149.0000	75.0000	118.9859	138.4354
Std(deg)	0.0000	0.0000	21.9047	12.1543
Herfindahl–Hirschman Index (HHI)	0.0067	0.0133	0.1419	0.1182
Std(HHI)	0.0000	0.0000	0.1040	0.0881
n^{HHI}	149.0000	75.0000	9.7423	11.5994
$\varphi_{(2)}$	-0.0067	$\simeq 0$	0.5653	0.5553
$\varphi_{\min}$	-0.0067	-1	-0.5151	-0.4419
Density	1	0.5034	0.7560	0.9105

- Summary network statistics
  - Networks become more *connected and diversified* as the phases progress.
  - Networks deviate from a bipartite structure but continue to display heterogeneity that differs from the linear-in-means structure.
  - A persistent core-periphery structure across all phases is observed.

Phase	1	2
λ_d (same exporter)	0.3002***	0.1370*
	(0.0362)	(0.0712)
λ_o (same importer)	0.3510***	0.2830***
	(0.0392)	(0.0649)
λ_w (third-party)	0.3354***	0.5734***
	(0.0336)	(0.0418)
Distance	-1.5184***	-1.7511***
	(0.2449)	(0.2849)
Border	0.9973***	0.9402***
	(0.2134)	(0.1575)
FTA	0.4123***	0.4968***
	(0.1717)	(0.0851)
McFadden's R^2	0.2848	0.1739

Note: Significance levels: *** (1%), ** (5%), * (10%). Standard errors are in parentheses. McFadden's $R^2 = 1 - \frac{\widehat{\ell}_N(\text{=our model})}{\widehat{\ell}_N^{trad.}(\text{traditional gravity model})}$

- λ_w is positive and remains highly significant.
  - persistent and significant *third-party* network effects.
- **Phase 1 (1997, active NAFTA implementation):** All interaction parameters turn positive,
  - reflecting a structural shift toward *complementarity*.
  - Regional integration under NAFTA
    - reduced trade frictions
    - deepened production linkages across member countries
  - capture the emergence of regional value chains and increasing returns to network connectivity.
- **Phase 2 (2016, expansion of global value chains):** λ_d and λ_o remain positive,
  - as the value of participating in the network increases when more countries become interconnected under global value chains,
  - reinforcing complementarities across the system.

5. Empirical Application

Counterfactual analysis

- Counterfactual scenario: Focusing on Phase 2, we consider a threefold increase in $\pi^{+}_{\mathrm{US,CN}}$ and $\pi^{+}_{\mathrm{CN,US}}$, thereby illustrating the recent US-China trade war.
- We compute $\widehat{s}_{ij}$ and $\widetilde{s}_{ij}$: for each $i = 1, \cdots, n$,

$$\widehat{s}_{ij} = \frac{\widehat{\mu}_{ij}}{\sum_{k=1,k\neq i} \widehat{\mu}_{ik}} \text{ (estimated) and } \widetilde{s}_{ij} = \frac{\widetilde{\mu}_{ij}}{\sum_{k=1,k\neq i}^{n} \widetilde{\mu}_{ik}} \text{ (counterfactual) for } j \neq i.$$

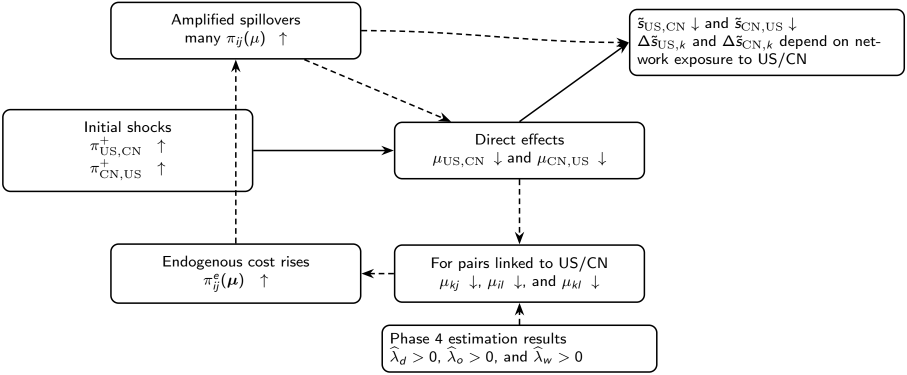
  - For comparison, we also compute $\widehat{s}_{ij}^{\mathrm{con}}$ and $\widetilde{s}_{ij}^{\mathrm{con}}$ from the conventional model.

- Shock: $\pi^{+}_{\mathrm{US,CN}} \uparrow$, $\pi^{+}_{\mathrm{CN,US}} \uparrow$
- Conventional gravity model ($\lambda = 0$)
  - Direct: $\mu_{\mathrm{US,CN}},\ \mu_{\mathrm{CN,US}} \downarrow$
  - Indirect
    - ⋆ MR only: $\{P_i, \Pi_j\}$ adjust $\Rightarrow$ *limited* third-country effects
- Our model (Phase 2: $\hat{\lambda}_d, \hat{\lambda}_o, \hat{\lambda}_w > 0$)
  - Direct: $\mu_{\mathrm{US,CN}},\ \mu_{\mathrm{CN,US}} \downarrow$
  - Indirect
    - ⋆ Endogenous network cost rises for pairs linked to US/CN: $\widetilde{\pi}^{e}_{ij}(\boldsymbol{\mu}; k, l) \uparrow$
    - ⋆ Amplified propagation: $\pi(\boldsymbol{\mu}) \Rightarrow \{P_i(\boldsymbol{\mu}), \Pi_j(\boldsymbol{\mu})\}$
    - $\Rightarrow$ Large, system-wide reallocation of import shares

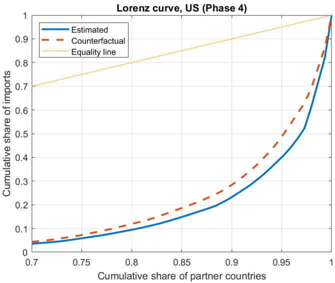
5. Empirical Application

Counterfactual analysis: Model mechanism

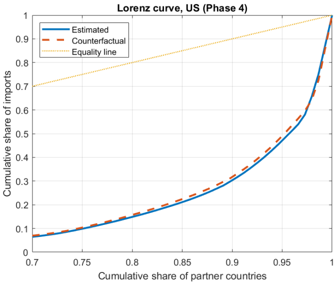
- Model's mechanism
  - Solid arrows: channels that are present in the conventional model
  - Dashed arrows: additional propagation mechanisms implied by $\widehat{\lambda} > 0$



(a) U.S. (Our model)



(b) U.S. (Conventional)

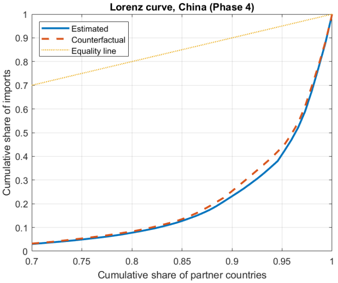


- U.S.-China shock $\Rightarrow$ U.S. import shares strongly reallocated toward many alternative suppliers.
  - Our model: Loss absorbed by many partners, Import concentration $\downarrow$
  - Conventional gravity: Adjustment concentrated on few large suppliers.

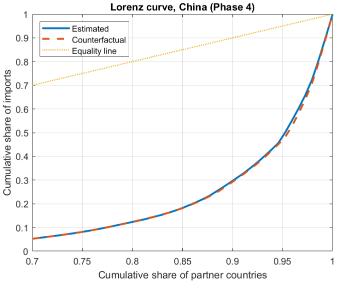
5. Empirical Application

Counterfactual analysis

(a) China (Our model)



(b) China (Conventional)

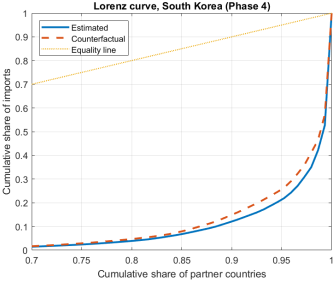


- U.S.-China shock
  - Our model: China experiences diversification under spillovers.
  - Conventional gravity: Minimal third-country adjustment

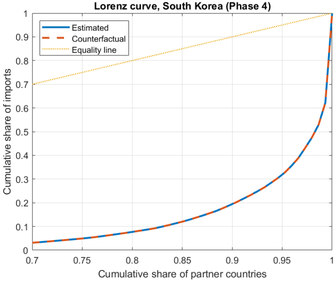
5. Empirical Application

Counterfactual analysis

(a) South Korea (Our model)



(b) South Korea (Conventional)



- U.S.-China shock $\Rightarrow$ Non-targeted countries respond through network exposure, not bilateral costs.

5. Empirical Application

Counterfactual analysis: Overall results and policy implications

- Takeaway
  - Trade policy shocks reshape the *global trade structure* through networks.
    - ★ Network-based endogenous trade costs generate large third-country adjustments and distributional responses.
    - ★ These are largely absent in the conventional gravity models.
  - Targeted hub economies (U.S., China):
    - ★ Import shares reallocated toward many partners ⇒ Diversification (lower concentration)
  - Third countries (e.g. Korea):
    - ★ Large reallocation despite no direct bilateral shock ⇒ Network exposure matters
    - ★ Spillovers occur even when π_{ij}^{+} is unchanged
- **Policy implications**: Ignoring network spillovers leads to a systematic understatement of trade policy effects.
  - Network-aware policy evaluation
    - ★ Bilateral tariffs generate global distributional effects
  - Hub countries matter
    - ★ Policies targeting hubs propagate disproportionately

Conclusion

- Research question: How do countries operate trade costs leveraging their trade networks?
- To this end, we characterized a network-interaction-based gravity equation.
- For estimation, we developed a Poisson pseudo maximum likelihood estimator with network effects and relevant tools for statistical inference.
- In empirical application, we expect to find evidence of endogenous trade costs, followed by implications.
- Counterfactual experiments illustrate how spillovers reshape global trade structures.